

Ps and Qs: Quantization-Aware Pruning for Efficient Low Latency Neural Network Inference

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ABSTRACT

Efficient machine learning implementations optimized for inference in hardware have wide-ranging benefits, depending on the application, from lower inference latency to higher data throughput and reduced energy consumption. Two popular techniques for reducing computation in neural networks are pruning, removing insignificant synapses, and quantization, reducing the precision of the calculations. In this work, we explore the interplay between pruning and quantization during the training of neural networks for ultra low latency applications targeting high energy physics use cases. Techniques developed for this study have potential applications across many other domains. We study various configurations of pruning during quantization-aware training, which we term *quantization-aware pruning*, and the effect of techniques like regularization, batch normalization, and different pruning schemes on performance, computational complexity, and information content metrics. We find that quantization-aware pruning yields more computationally efficient models than either pruning or quantization alone for our task. Further, quantization-aware pruning typically performs similar to or better in terms of computational efficiency compared to other neural architecture search techniques like Bayesian optimization. Surprisingly, while

networks with different training configurations can have similar performance for the benchmark application, the information content in the network can vary significantly, affecting its generalizability.

Keywords: pruning, quantization, neural networks, generalizability, regularization, batch normalization

1 INTRODUCTION

Efficient implementations of machine learning (ML) algorithms provide a number of advantages for data processing both on edge devices and at massive data centers. These include reducing the latency of neural network (NN) inference, increasing the throughput, and reducing power consumption or other hardware resources like memory. During the ML algorithm design stage, the computational burden of NN inference can be reduced by eliminating nonessential calculations through a modified training procedure. In this paper, we study efficient NN design for an ultra-low latency, resource-constrained particle physics application. The classification task is to identify radiation patterns that arise from different elementary particles at sub-microsecond latency. While our application domain emphasizes low latency, the generic techniques we develop are broadly applicable.

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